

Sequences

Problem 1. Define sequence $(a_n)_{n=1}^{\infty}$ by $a_1 = a_2 = a_3 = 1$ and $a_{n+3} = a_{n+1} + a_n$ for all $n \geq 1$. Also, define sequence $(b_n)_{n=1}^{\infty}$ by $b_1 = b_2 = b_3 = b_4 = b_5 = 1$ and $b_{n+5} = b_{n+4} + b_n$ for all $n \geq 1$. Prove that there exists $N \in \mathbb{N}$ such that $a_n = b_{n+1} + b_{n-8}$ for all $n \geq N$.

Problem 2. Let $a_1, a_2, a_3, \dots$ be an infinite sequence of nonnegative real numbers such that for all positive integers k the following conditions hold:

(i) $a_k - 2a_{k+1} + a_{k+2} \geq 0$;

(ii) $\sum_{j=1}^k a_j \leq 1$.

Prove that for all positive integer k the inequality $0 \leq a_k - a_{k+1} < \frac{2}{k^2}$ holds.

Problem 3. Let $a_1, a_2, \dots$ be an infinite sequence of integers that satisfies $a_{n+2} = a_{n+1} + a_n$ for all $n \geq 1$. There exists a positive integer k such that $a_k = a_{k+2018}$. Prove that there exists a term of the sequence which is equal to zero.

Problem 4. Let $n \geq 0$ be an integer. A sequence $a_0, a_1, a_2, \dots$ of integers is defined as follows: we have $a_0 = n$ and for $k \geq 1$, a_k is the smallest integer greater than a_{k-1} for which $a_k + a_{k-1}$ is the square of an integer. Prove that there are exactly $\lfloor \sqrt{2n} \rfloor$ positive integers that cannot be written in the form $a_k - a_\ell$ with $k > \ell \geq 0$.

Problem 5. Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence of real numbers such that

$$a_{n+2} = |a_{n+1}| - a_n$$

for all $n \geq 1$. Prove that there exists $d \in \mathbb{N}$ such that $a_{n+d} = a_n$ for all $n \in \mathbb{N}$.

Problem 6. Suppose we colour every positive integer red or blue randomly. Prove that there exists an infinite sequence of positive integers $a_1 < a_2 < \dots$ such that the infinite sequence of positive integers $a_1, \frac{a_1 + a_2}{2}, a_2, \frac{a_2 + a_3}{2}, a_3, \dots$ has the same colour.

Problem 7. Let $a_1, a_2, a_3, \dots$ be the sequence of real numbers defined by $a_1 = 1$ and

$$a_m = \frac{1}{a_1^2 + a_2^2 + \dots + a_{m-1}^2} \quad \text{for } m \geq 2.$$

Determine whether there exists a positive integer N such that

$$a_1 + a_2 + \dots + a_N > 2017^{2017}.$$

Problem 8. Let α be a real number in the interval $(0, 1)$. Prove that there exists a sequence $(\varepsilon_n)_{n \geq 1}$ where each term is either 0 or 1 such that the sequence $(s_n)_{n \geq 1}$ defined by

$$s_n = \frac{\varepsilon_1}{n(n+1)} + \frac{\varepsilon_2}{(n+1)(n+2)} + \dots + \frac{\varepsilon_n}{(2n-1)2n}$$

satisfies the inequality

$$0 \leq \alpha - 2ns_n \leq \frac{2}{n+1}$$

for any $n \geq 2$.